

Ash-free fiber (aNDFom): Why the change and what are the consequences?

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Ash-free fiber (aNDFom): Why the change and what are the consequences?

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Introduction

- For ruminants, fiber in plants should contain the cellulose, hemicellulose and lignin in cell walls
 - Pectin in cell walls is easily fermented
 - Digestible cellulose and hemicellulose are organic structural carbohydrates that contribute energy to ruminants via VFAs
 - Neutral detergent fiber is the best measure of total insoluble fiber for ruminants

Introduction

- Fiber is a nutritional entity
 - The indigestible or slowly digesting fraction of a feed that occupies space in the gut
 - Differs with the type of animal
- Fiber is not a chemical entity
 - We attempt to use chemical extraction to measure the nutritional entity (fiber)
 - Analytical methods for fiber are empirical
 - The method defines the fiber fraction measured
 - There are compromises to make methods effective and efficient

Introduction

- Depending on the method used, NDF contains some contaminants
 - aNDF (AOAC Official Method) used amylase and sulfite and was isolated without ashing
 - Contaminated with less CP and some ash
 - Rapid method because there was no ashing step
 - aNDFom (AOAC Official Method) used amylase and sulfite, and is ash-free
 - Contaminated with some CP
 - Only the organic matter in fiber contributes energy to the animal

Introduction

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Introduction

- Depending on the method used, NDF contains some contaminants
 - aNDFom (AOAC Official Method) used amylase and sulfite, and is ash-free
 - aNDF and aNDFom will result in less aNDICP, so separate NIR calibrations are needed
 - Ashing crucibles typically results in loss of weight from empty crucibles, so a blank correction after ashing is needed to accurately measure aNDFom

Introduction

- Methods and their acronyms are important
 - NDF
 - NDR
 - aNDF
 - aNDFom
 - They are all a measure of fiber, but not the same measure of fiber!
- Everyone has to be more accurate in fiber measurement, reporting and usage
 - My patience is wearing thin!

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Introduction

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Why worry about NDF contamination? Crude Protein Contamination

Feed	CP	NDR	NDICP (w/o sulfite)	aNDF	NDICP (w/ sulfite)
Fish meal	53.94	30.44	10.43	6.27	1.29
Brewer's grains	30.44	52.32	12.16	40.87	4.65
Distiller's grains	25.57	38.58	11.01	27.89	3.68
Soybean meal	46.15	18.48	3.63	12.44	0.48
Sunflower meal	31.86	38.52	2.38	35.20	1.14
Canola meal	40.83	23.73	4.33	20.88	2.09
Citrus pulp	6.53	21.27	2.06	20.20	1.59

**There can be a huge difference in NDF and NDICP between the NDR and aNDF methods
CANNOT use NDICP measured by NDR to adjust aNDF for protein contamination**

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Why worry about NDF contamination? Crude Protein Contamination

Feed	CP	NDR	NDICP (w/o sulfite)	aNDF	NDICP (w/ sulfite)
Grasses	8.69	66.82	2.32	65.02	1.73
Corn silage	7.65	36.08	0.72	34.74	0.50
Legumes	17.32	40.32	2.71	38.91	1.65

CP contamination of forages is small especially for the aNDF method, which uses sulfite

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Why worry about NDF contamination? CP versus Ash Contamination

Feed	NDICP (w/ sulfite)	aNDF (w/ sulfite)	aNDFom (w/ sulfite)	aNDF ash (w/ sulfite)
Grasses	1.73	55.60	53.64	1.96
Corn silage	0.50	36.31	35.36	0.95
Legumes	1.65	40.45	38.40	2.05

For the aNDF method, the correction for ash contamination is typically greater than that for CP contamination

If we need to "correct" aNDF for NDICP, then we also need "correct" for aNDF ash to more accurately measure fiber for models, summative equations, and the calculation of NFC

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Why worry about NDF contamination? CP versus Ash Contamination

Feed	NDICP (w/ sulfite)	aNDF (w/ sulfite)	aNDFom (w/ sulfite)	aNDF ash (w/ sulfite)
Grasses	1.73	55.60	53.64	1.96
Corn silage	0.50	36.31	35.36	0.95
Legumes	1.65	40.45	38.40	2.05

For routine use, the correction for NDFash is more important than correction for NDICP because fiber-containing feeds are typically fed to provide energy, which can only be obtained from organic matter (OM)

Correct measurement and partitioning of OM is crucial, and soil contamination of aNDF can be large

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Why be concerned about ash contamination of aNDF?

- Conceptually, OM should be the basis for energy evaluation of feeds
 - Only OM can be used to generate energy
 - Europeans use digestible OM (dOM or “d value”) to measure the energy potential of feeds
 - US used TDN, which was also ash-free
 - $TDN = dCP + dCF + dNFE + 2.25 \cdot dEE$
 - Digestible DM (dDM) is correlated with dOM but is affected by digestible ash

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Why be concerned about ash contamination of aNDF?

- Results in over estimation of fiber in the diet
 - Crucial when low fiber dairy rations are fed
 - Example, dairy farmer fed 30% of ration DM as alfalfa silage containing 36% aNDF, but experienced milk fat depression.
 - The aNDFom of the alfalfa was only 28%, so the pNDF of the ration was below minimum requirements

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What is Ash Contamination of aNDF

- Ash intrinsic to fiber
 - Ash that is a part of fiber (silica) or ash that is not extracted by ND
 - Typically 15 to 25% of total ash in forages
 - Corn silage = 3-5% ash
 - Grasses = 6-8% ash
 - Legumes = 8-10% ash
- Ash from soil contamination 0-10% of DM
 - Raking, splashing during rains, flooding
 - Gravel pads for silage or hays
 - Soil is insoluble in ND

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Why be concerned about ash contamination of aNDF?

- Results in over estimation of fiber in the diet
 - Can cause a significant under estimation of NFC in feeds
 - $NFC = 100 - \text{ash} - CP - \text{Fat} - \text{aNDF}$
 - Example (alfalfa with soil contamination):
 - $NFC = 100 - 17 - 20 - 3 - 36 = 24$
 - $NFC = 100 - 17 - 20 - 3 - 28 = 32$
 - The error in NFC creates an under estimation of TDN or NE when summative equations are used because true digestibility of NFC is 0.98, which is more than the digestibility of aNDF

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Why be concerned about ash contamination of aNDF?

- Need to detect soil contamination and accurately measure aNDFom
- Need to be more accurate in the calculation of NFC
- Need to accurately calculate usable energy in feeds (TDN and NE)

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Why be concerned about ash contamination of aNDF?

- $TDN_{1X} = tdCP + tdFA \cdot 2.25 + tdNDF + tdNFC - 7$
(2001 Nutrient Requirements of Dairy Cattle)
 - $tdCP(\text{for}) = CP \times \exp(-1.2 \times ADICP/CP)$
 - $tdCP(\text{conc}) = CP \times (1 - 0.4 \times ADICP/CP)$
 - $tdFA = (EE - 1) \times 1.00$
 - $tdNDF = (NDF - NDICP - \text{Lignin}) \times .75 \times [1 - (\text{Lignin}/(NDF - NDICP))^{0.667}]$
 - OR $tdNDF = IVNDFomD \cdot aNDFom$
 - If aNDFom is measured then IVNDFomD must also be measured
 - Lignin equation is inappropriate for aNDFom

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Why be concerned about ash contamination of aNDF?

- Over estimation of fiber causes inaccurate estimates of TDN and NEL

Ash	NDR	aNDF	aNDF _{om}	aNDF _{ash}	NDICP	aNDICP	NFC	IVNDFD	IVNDF _{omD}	TDN
10.0	40.0				3.0		31.0	0.377		58.9
10.0		38.6				2.0	31.4	0.391		59.3
10.0			36.6	2.0		2.0	33.4	0.412		61.2
15.0	45.0				3.0		21.0		0.335	49.1
15.0		43.6				2.0	21.4		0.346	49.5
15.0			36.6	7.0		2.0	28.4		0.412	56.3

For normal samples, using aNDF_{om} increases NFC and TDN slightly
For soil contaminated samples, not using aNDF_{om} reduces NFC and TDN dramatically

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Why be concerned about ash contamination of aNDF?

- Results in over estimation of fiber in the diet
 - Paradoxically, aNDF generates better estimates of NE when regression equations are used
 - Grasses: $NEL = 2.860 - .0262 \cdot NDF$
 - Legumes: $NEL = 2.323 - .0216 \cdot NDF$
 - These equations SHOULD NOT be used with aNDF_{om}
 - They imply that the lower NE is due to fiber, when it is actually lower due to ash
 - Summative equations not only are more accurate when aNDF_{om} is used but also clearly indicates that the problem is ash and its impact on NFC

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Why be concerned about ash contamination of aNDF?

- Results in over estimation of fiber in the diet
 - Paradoxically aNDF generates better estimates of NE when regression equations are used
 - Grasses: $NEL = 2.860 - .0262 \cdot NDF$
 - Legumes: $NEL = 2.323 - .0216 \cdot NDF$
 - Example (soil contamination):
 - $NEL = 2.323 - .0216 \cdot 36 = 1.545$ Mcal NE/kg DM
 - $NEL = 2.323 - .0216 \cdot 28 = 1.761$ Mcal NE/kg DM
 - A soil-contaminated sample should have less NE because the added ash generates no NE, but using aNDF_{om} results in a higher estimate of NE
 - In this situation aNDF gives a better estimate of NE than does aNDF_{om}, but for the wrong reasons.

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What are the consequences of using aNDF_{om}?

- NFC, OM partitioning and energy estimation are more accurate if aNDF_{om} is used
- Regression equations developed using NDF to predict TDN or NE cannot be used with aNDF_{om}

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Why be concerned about ash contamination of aNDF?

- Results in over estimation of fiber in the diet
 - Paradoxically, aNDF generates better estimates of NE when regression equations are used
 - Grasses: $NEL = 2.860 - .0262 \cdot NDF$
 - Legumes: $NEL = 2.323 - .0216 \cdot NDF$
 - These equations were generated from research samples that had little or no soil contamination
 - Both the intercept and regression coefficients were generated assuming typical intrinsic ash in NDF

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What are the consequences of using aNDF_{om}?

- For most forages that are not heated, "correction" of aNDF_{om} for NDICP (using sulfite) is not needed to adequately measure NFC and energy value
 - The assumption that all N is protein by multiplying $N \times 6.25$ is probably a larger error in summative equations than is aNDICP contamination

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What are the consequences of using aNDFom?

- If aNDFom is used, the minimum peNDF and optimum aNDF recommendations should be reduced to 95%
 - Most (all?) recommendations for NDF were based on NDF, NDF or aNDF measurements that were not corrected for ash

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Questions ?

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Conclusions

- aNDFom is the most accurate measure of fiber that can generate energy for the animal
- When using aNDFom summative equations must be used to estimate TDN or NE
- When using aNDFom in summative equations IVNDFomD must be used to estimate dNDFom

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Conclusions

- aNDFom should not be substituted into regression equations developed for predicting TDN or NE using NDF
- Recommendations for optimal NDFom and minimum peNDFom should be about .95 of the current recommendations for optimal NDF and minimum peNDF

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